

Accelerated Aging: Progeria

Human progeria comes in two major forms, Werner's syndrome (adult-onset progeria) and Hutchinson-Gilford syndrome (juvenile-onset progeria). Werner's patients are usually diagnosed in early maturity and have an average life span of forty-seven years. Hutchinson-Gilford patients are usually diagnosed within the first two years of life and have an average life span of thirteen years. The latter syndrome is often simply termed "progeria" and both are sometimes lumped together as progeroid syndromes.

Progeria's Effects

There is considerable controversy as to whether or not progeria is a form of aging at all. Most clinicians believe that progeria is truly a form of early aging, although only a segmental form in which only certain specific tissues and cell types of the body age early. Hutchinson-Gilford children show what appears to be early aging of their skin, bones, joints, and cardiovascular system, but not of their immune or central nervous systems.

Clinical problems parallel this observation: They suffer from thin skin and poor skin healing, **osteoporosis**, arthritis, and heart disease, but do not have more infections than normal children and they do not have early **dementia**. Death is usually due to cardiovascular disease, especially heart attacks and strokes, yet Hutchinson-Gilford children lack normal risk factors associated with these diseases, such as smoking, high cholesterol, hypertension, or diabetes.

Clinically, the children appear old, with thin skin, baldness, swollen joints, and short stature. They do not go through puberty. The face is strikingly old in appearance. The typical Hutchinson-Gilford child looks more like a **centenarian** than like other children, and may look more like other progeric children than like members of their own families. There is no effective clinical intervention.

Inheritance of Progeria

The segmental nature of progeria is perhaps its most fascinating feature. If progeria is actually a form of aging gone awry, then this implies that aging is more than merely wear and tear on the organism. If progeria is a genetically mediated, segmental form of aging, this may imply that aging itself is



osteoporosis thinning of the bone structure

dementia neurological illness characterized by impaired thought or awareness

centenarian person who lives to age 100

This five year old boy has Hutchinson-Gilford progeria, a fatal, "premature aging" disease in which children die of heart failure or stroke at an average age of thirteen. Photo courtesy of The Progeria Research Foundation, Inc. and the Barnett Family.



genetically mediated and, like other genetic disease, is not only the outcome of genetic error but might be open to clinical intervention.

Supporting this observation, there are a number of other less well-known forms of progeria, including acrogeria, metageria, and acrometageria, as well as several dozen human clinical syndromes and diseases with features that have been considered to have progeroid aspects. The latter category includes Wiedemann-Rautenstrauch, Donohue's, Cockayne's, Klinefelter's, Seip's, Rothmund's, Bloom's, and Turner's syndromes, ataxia telangiectasia, cervical lipodysplasia, myotonic dystrophy, dyskeratosis congenita, and trisomy 21 (Down syndrome). In each of these cases, there are features that are genetic and that have been considered segmental forms of aging.

In the most well-known of these, trisomy 21, the immune and central nervous systems both appear to **senesce** early, in contrast with Hutchinson-Gilford progeria, in which the opposite occurs. Bolstering the suggestion that this is a form of segmental progeria, trisomy 21 patients are prone to both infections and early onset of a form of Alzheimer's dementia.

The gene that is mutated in Werner's syndrome is known to code for a DNA helicase. This enzyme unwinds DNA for replication, transcription, recombination, and repair. The inability to repair DNA may explain the features of premature aging, as well as the increased rate of cancer in Werner's syndrome patients. Another mutated helicase is responsible for Bloom's syndrome. Both conditions are inherited as **autosomal** recessive disorders.

Data suggesting that Hutchinson-Gilford progeria is genetic is circumstantial. The disease is presumptively caused by a sporadic (one in eight million live births), autosomal dominant mutation, although a rare autosomal recessive mutation is not impossible. The helicase abnormality that causes Werner's syndrome is not present in Hutchinson-Gilford cells. There is a slight correlation with the paternal age at conception. Whatever the mechanism, it appears to operate prior to birth; several **neonatal** cases have been reported.

Germinal Mosaicism

Cellular data, particularly regarding structures called telomeres, suggests that some of the cells from Hutchinson-Gilford patients are prone to early cell senescence. Telomeres are special DNA structures at the tips of the chromosomes. These telomeres gradually shorten over time, and this shortening is associated with some aspects of cellular aging. Skin fibroblasts from Hutchinson-Gilford patients have shorter than normal telomeres and consequently undergo early cell senescence. At birth, the mean telomere length of these children is equivalent to that of a normal eighty-five-year-old.

Introduction of human telomerase into such cells leads to reextension of the telomeres and results in normal immortalization of these progeric cell cultures. Clinical interventional studies using this strategy in humans are pending. Predictably, circulating **lymphocytes** of Hutchinson-Gilford children have normal telomere lengths, in keeping with their normal immune function. Research thus far suggests that progeria may not be so much a genetic disease as it is an "epigenetic mosaic disease." In progeria, this means that the genes are normal, but the abnormally short telomere length in only certain cell lines causes an abnormal pattern of gene expression. The senescent pattern of gene expression in specific tissues results in the observed clinical disease of progeria.

Although consistent with all known laboratory and clinical data, the actual genetic mechanisms that underlie Hutchinson-Gilford progeria are still uncertain and arguable (the gene for Werner's syndrome, however, has been cloned). The question of what causes progeria holds a fascination largely for what it may tell us about the course of aging itself. SEE ALSO AGING AND LIFE SPAN; ALZHEIMER'S DISEASE; DISEASE, GENETICS OF; DNA REPAIR; DOWN SYNDROME; INHERITANCE PATTERNS; MOSAICISM; TELOMERE.

senesce a state in a cell in which it will not divide again, even in the presence of growth factors

autosomal describes a chromosome other than the X and Y sex-determining chromosomes

neonatal newborn

lymphocytes white blood cells



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Addiction

Addiction in its broadest sense can be defined as the habituation to a practice considered harmful. A more narrow definition of the term refers to chronic use of a chemical substance in spite of severe psychosocial consequences. Terms such as “workaholic,” “sex addict,” and “computer junkie” arose to describe behaviors that have features in common with alcoholism and other substance addictions. The most convincing data supporting a role of genetics in addiction has been collected for alcoholism, although genetics most likely has a role in other forms of addiction.

Definitions

In order to assess alcoholism, or any form of addiction, a clear definition of the condition is necessary. The American Psychiatric Association and the World Health Organization have developed clinical criteria (DSM-IV and ICD10, respectively) that are widely used for the diagnosis of substance-use related disorders. DSM-IV criteria recognizes ten classes of substances (alcohol, amphetamines, cannabis, hallucinogens, inhalants, nicotine, opioids, phencyclidine, and sedatives) that lead to substance dependence, another term for addiction.

The precise diagnostic criteria for dependence vary among substances. DSM-IV defines dependence as manifesting, within a twelve-month period, at least three of the following criteria:

- Tolerance (increased dose needed to achieve the same affect, or reduced response to the same dose)
- Withdrawal symptoms
- Progressive increase in dose or time used
- Persistent desire for, or failure to reduce substance use
- Increasing efforts made to obtain substance
- Social, occupational, or recreational activity is replaced by activity associated with substance use
- Continued substance use despite recognized physical and psychological consequences

Heritability in Humans

Most family, twin, and adoption studies have shown that addiction to alcohol has significant heritability. For example, there is an increased risk for alcoholism in the relatives of alcoholics. Depending on the study, the risk of alcoholism in siblings of alcoholics is between 1.5 and 4 times the risk for the general population. The identical twins of alcoholics (who share 100

percent of their genes) are more likely to be alcoholics than the fraternal twins of alcoholics (who share only about 50 percent). Adoption study data suggest that the risk for developing alcoholism for adopted children is influenced more by whether their biological parents were alcoholics than whether their adopted parents are alcoholics, suggesting that genes contribute to alcoholism more than environment. Similar but less extensive data has been collected for nicotine addiction. Very little genetic epidemiological data has been collected for illegal drugs.

The only genes that have been conclusively shown to affect susceptibility to addiction in humans are genes that encode proteins responsible for the metabolism of alcohol. In the body, ethanol (“drinking” alcohol) is oxidized by enzymes to acetaldehyde and then to acetate. Certain **alleles** of aldehyde dehydrogenase genes that are common in some populations, such as Asians, lead to increased levels of acetaldehyde when alcohol is consumed. Acetaldehyde causes an unpleasant flushing reaction that leads to a voluntary reduction of alcohol consumption. The systematic search for other genes that affect susceptibility to alcohol and nicotine addiction in humans has led to the identification of chromosome **loci** that may contain genes that affect susceptibility to addiction, but has not led to the identification of any specific genes.

Models of Addiction

Progress in genetic analysis of addiction in animal models has been more successful. The pharmacologic effects of abused substances can readily be demonstrated in many model systems, from worms to rodents. Rodents can be trained to voluntarily consume alcohol and other abused substances. Once trained, these rodents will expend energy to continue to receive drugs and will display withdrawal symptoms when denied drugs. Chromosomal regions with naturally occurring variants that affect voluntary consumption, intoxication, and withdrawal have been mapped in mice. The specific genes responsible for these effects have not yet been identified.

Cell biology and neurochemistry studies in humans and model systems have identified many molecules that have altered abundance and distribution, enzymes with altered activity, and genes with altered expression resulting from substance abuse. In particular, the dopamine and serotonin **neurotransmitter** systems have been the focus of intense studies. These are brain systems directly involved in many basic responses, including pleasure and reward systems.

To directly test the role of specific genes and pathways, mice have been engineered to delete or over-express genes. Mice lacking any of these genes (called PKC_ε, DRD2, and DBH) are more sensitive to the effects of alcohol and consume less alcohol. In contrast, mice lacking any one of four other genes (PKA regulatory II_β, NPY, or 5-HT_{1b}) are less sensitive to the effects of alcohol and consume more alcohol. Mice cannot be trained to self-administer alcohol if they lack the Mu opioid receptor, which is involved in transmitting signals to the body’s own internal opiate system.

Mutant fruit flies with altered responses to alcohol intoxication have also been created. Two mutants, called “cheapdate” and “amnesiac,” arise from different mutations in the same gene. These mutations affect the cellular level of the signal transduction molecule cyclic-AMP. As the names imply,

alleles particular forms of genes

loci sites on a chromosome (singular, locus)

neurotransmitter molecule released by one neuron to stimulate or inhibit a neuron or other cell

flies with cheapdate mutations are very sensitive to the affects of alcohol, and flies with amnesiac mutations are unable to learn.

The major conclusion from work in model systems is that the pathways and systems involved in addiction are central to normal behaviors with instinctive reward processes, such as feeding and procreation. Addiction is a process that involves learning and the subversion of these basic reward pathways. SEE ALSO COMPLEX TRAITS; DISEASE, GENETICS OF; GENE AND ENVIRONMENT; SIGNAL TRANSDUCTION; TWINS.

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Aging and Life Span

Aging is, simply put, the act of getting older. Aging is part of the natural life cycle of an organism. From birth, through maturation, and eventually to death, aging is the element that ties all segments of life together.

Life Span and the Aging Process

How long an organism lives is called its life span. In 1998, the average life span for a human, worldwide, was sixty-six years. However, life span is a complex trait, meaning that many factors, including family history, lifestyle, disease, and residence in a developed nation, determine how long an individual's life will be. The average life span in a particular population changes as these factors change. For example, the average life span in the United States in 1900 was forty-nine; in 1998 it was seventy-seven.

This increase was likely due to several factors, but perhaps the most important was the improvement of sanitation, hygiene, and public health from 1900 to 1998. These improvements included purification of drinking water, treatment of wastewater, widespread vaccination, and improved access to health care. However, even as these sanitary measures were adopted, other elements of modern life emerged as strong influences on life span, such as diet, exercise, and socioeconomic status. Studies have shown that individuals who exercise regularly, eat a diet lower in saturated fats, and avoid unnecessary risk-taking live longer. This may be because such a lifestyle reduces the risk of developing cardiovascular disease and cancer, the top causes of death in developed countries.

Finally, life span is in part genetically determined. Studies of life span in large families have shown that longevity is, to some degree, inherited. This may be due to shared genetic risks of diseases or behaviors that shorten

the life span, or it may reflect direct genetic influences in longevity separate from risk of disease.

The aging process causes many changes, both visible and invisible. In humans, these changes take several forms. In the first two decades of life, from birth to adulthood, aging involves physical growth and maturation and intellectual development. These changes are fairly noticeable and relatively swift compared to the rest of the life span. After reaching physical maturity, humans begin to show subtle signs of physical aging that grow more pronounced over time. Long-term exposure to sunlight and the outdoors may begin to toughen the skin and produce wrinkles on the face and body. The senses change: Sight, hearing, taste, and smell become less acute. Gradual changes in the eye cause many older adults to need glasses to read. Hair begins to thin and turn gray. Individuals with less active lifestyles often begin to gain weight, particularly around the waist and hips. Beginning in their 40s (or, rarely, in their late 30s), many women experience menopause, which marks the end of childbearing years. Less visible or noticeable changes associated with aging are the loss of bone density over time (particularly in women), slower reflexes, less acute mental agility, and declining memory.

Diseases Associated with Aging

Many of the diseases common in older adults, such as cardiovascular disease, cancer, **dementia**, arthritis, blindness, and deafness, are consequences of the acceleration or distortion of these “natural” changes associated with aging. These complex diseases associated with aging are caused by the interaction of genetic and environmental factors. For example, Alzheimer’s disease is an illness caused by changes in the brain that impair thinking and memory much more severely than the natural decline that all humans experience during aging. It is the most common form of dementia in adults over age sixty. Certain very rare **alleles** are associated with the development of Alzheimer’s disease, and other much more common alleles (of different genes) increase the risk of a variety of other diseases. Environmental factors such as exposure to toxins have also been implicated. A more rare example is progeria, a disease in which the tissues of the body age about seven times more rapidly than normal. In this case, a person who is chronologically only a teenager looks much older.

Genetics and Aging

Many scientists have hypothesized that some genes may control aspects of aging separate from the development of disease. These hypotheses are based on experimental studies of non-human organisms and the observation that longevity in humans appears to run in families. Studies of yeast and roundworm have identified over ten genes in each that are associated with longevity and aging, and more recent studies have suggested similar genes exist in the fruit fly. The exact function of these genes is unknown, but one or more may help slow down the metabolic rate. Studies in mice have shown that reducing metabolism by reducing food intake can increase life span. Finally, shortening of the **telomeres** decreases longevity in some model organisms.

Finding similar genes in humans is more complicated, since scientists cannot experimentally control genes to test their effects on longevity in

dementia neurological illness characterized by impaired thought or awareness

alleles particular forms of genes

telomeres chromosome tips

Life span is determined partly by genetics: These grandchildren are more likely to live longer based on the long life span of their grandparents.



centenarians people who live to age 100

humans. Therefore, genetic studies of human longevity require a more observational approach. One study design is to examine large numbers of long-lived individuals such as **centenarians** and see what factors they have in common, such as lifestyle, medical history, and genetics.

Studies of centenarians have suggested that variants in multiple genes, including the human leukocyte antigen (HLA) genes of the immune system, apolipoprotein E (APOE), angiotensin-converting enzyme (ACE), plasminogen activating inhibitor 1 (PAI-1), and p53, are associated with living past age ninety. Forms of several of these genes, such as APOE, ACE, and p53, are associated with increased risk of developing Alzheimer's disease, cardiovascular disease, and cancer, respectively. The association of these genes with longevity may be due to these disease associations, or it may be due to their direct influence on extending the human life span. Regardless, genes clearly influence aging and longevity, whether it is by influencing the development of life span-shortening diseases, or by positively influencing longevity independently of causing disease. SEE ALSO ACCELERATED AGING; PROGERIA; ALZHEIMER'S DISEASE; CANCER; CARDIOVASCULAR DISEASE; COMPLEX TRAITS; TELOMERE.

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Agricultural Biotechnology

Biotechnology is the use of living organisms—microbes, plants, or animals—to provide useful new products or processes. In a broad sense, biotechnology continues a process that is thousands of years old. Using traditional plant breeding techniques, humans have altered the genetic composition of almost every crop by only planting seeds from plants with desired traits, or by controlling pollination. As a result, most commercial crops bear little resemblance to their early relatives. Current maize varieties are so changed from their wild progenitors that they cannot survive without continual human intervention.

The 1970s heralded recombinant DNA technology, which gave researchers the ability to cut and recombine DNA fragments from different sources to express new traits. Genes and traits previously unavailable through traditional breeding became available through DNA recombination.

Techniques

Modern plant genetic engineering involves transferring desired genes into the DNA of some plant cells and regenerating a whole plant from the transformed tissue. New DNA may be introduced into the cell via biological or physical means.

The most widely used biological method for transferring genes into plants capitalizes on a trait of a naturally occurring soil bacterium, *Agrobacterium tumefaciens*, which causes crown gall disease. This bacterium, in the course of its natural interaction with plants, has the ability to infect a plant cell and transfer a portion of its DNA into a plant's **genome**. This leads to an abnormal growth on the plant called a gall. Scientists take advantage of this natural transfer mechanism by first removing the disease-causing genes and then inserting a new beneficial gene into *A. tumefaciens*. The bacteria then transfer the new gene into the plant.

Another gene transfer technique involves using a "gene gun" to literally shoot DNA through plant cell walls and membranes to the cell nucleus, where the DNA can combine with the plant's own genome. In this technique, the DNA is made to adhere to microscopic gold or tungsten particles and is then propelled by a blast of pressurized helium.

Advantages

Depending on which genes are transferred, agricultural biotechnology can protect crops from disease, increase their yield, improve their nutritional content, or reduce pesticide use. In 2000, more than half of American soybeans and cotton and one-fourth of American corn crops were genetically

genome the total genetic material in a cell or organism





A golden rice field in Kaili, China. Golden rice is part of a “second generation” of genetically modified foods.

modified by modern biotechnology techniques. Genetically modified foods may also help people in developing countries. One in five people in the developing world do not have access to enough food to meet their basic nutritional needs. By enhancing the nutritional value of foods, biotechnology can help improve the quality of basic diets.

“Golden rice” is a form of rice engineered to contain increased amounts of vitamin A. Researchers are also developing rice and corn varieties with enriched protein contents, as well as soybean and canola oils with reduced saturated fat. Other potential benefits include crops that can withstand drought conditions or high salinity, allowing populations living in harsh regions to farm their land.

Agricultural biotechnology also provides benefits for the manufacture of pharmaceutical products. Because plants do not carry human diseases, plant-made vaccines and antibodies require less screening for bacterial toxins and viruses. In addition to plants, animals may also be engineered to produce beneficial genes. In order to produce large quantities of **monoclonal antibodies** for research on new therapeutic drugs, several compa-

monoclonal antibodies
immune system proteins derived from a single B cell

nies have genetically engineered cows and goats to secrete antibodies into their milk. One company has inserted a spider gene into dairy goats. The spider silk extracted from the goat's milk is expected to produce fibers for bulletproof vests and medical supplies, such as stitch thread, and other applications where flexible and extremely strong fibers are required.

Concerns

Despite the benefits of genetic engineering, there are concerns about whether recombinant DNA techniques carry greater risks than traditional breeding methods. Consumer acceptance of food derived from genetically engineered crops has been variable. Many individuals express concerns regarding the environmental impact and ethics of the new technology, and about food safety. One of the major food safety concerns is that there is a risk that crops expressing newly inserted genes may also contain new **allergens**.

Some groups have expressed concern that widespread use of plants engineered for specific types of pest resistance could accelerate the development of pesticide-resistant insects or have negative effects on organisms that are not crop pests. Another environmental concern is that transgenic, pest-protected plants could **hybridize** with neighboring wild relatives, creating "superweeds" or reducing genetic **biodiversity**.

Regulations

To address these concerns, agricultural biotechnology products are regulated by a combination of three federal agencies: the U.S. Department of Agriculture (USDA), the Environmental Protection Agency (EPA), and the Food and Drug Administration (FDA). Together, these agencies assess genetically modified crops, as well as products that use those crops. They test the crops and products for safety to humans and to the environment, and for their efficacy and quality. **SEE ALSO** BIOPESTICIDES; GENETICALLY MODIFIED FOODS; PLANT GENETIC ENGINEER; TRANSGENIC ANIMALS; TRANSGENIC MICROORGANISMS; TRANSGENIC PLANTS.

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allergens substances that trigger an allergic reaction

hybridize to combine two different species

biodiversity degree of variety of life

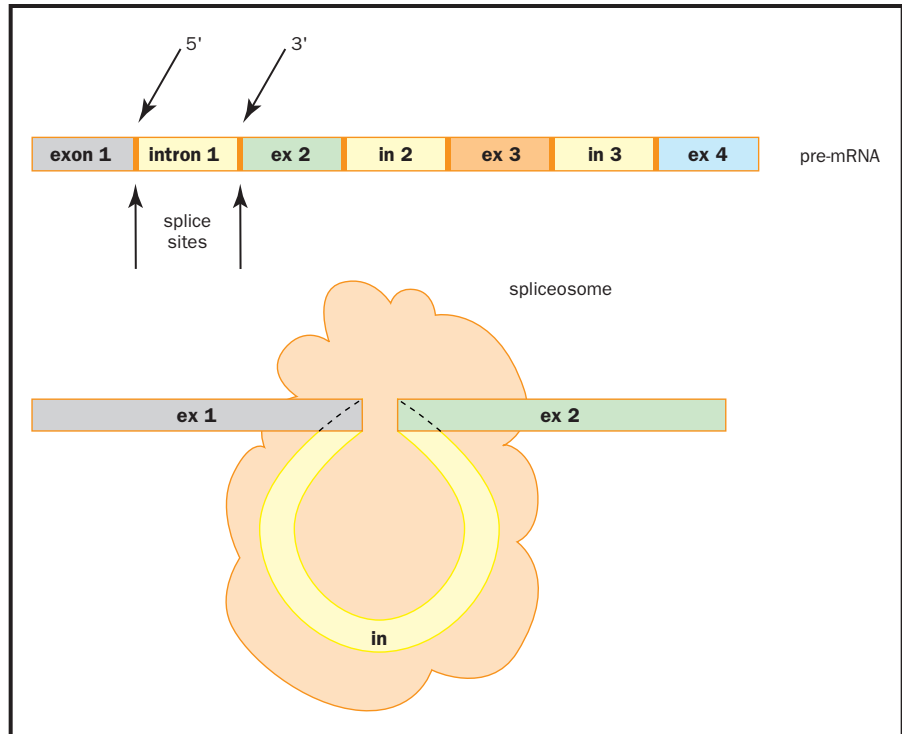
Alternative Splicing

When molecular biologists began analyzing the complete sequence of the human genome in mid-2001, one surprising observation was that humans have relatively few genes. We may have as few as 30,000 genes, only about two



Alternative Splicing

Introns are removed by the spliceosomes.



The sensitivity of the human ear to a wide range of sound frequencies is due to alternative splicing of a potassium channel gene, giving rise to a set of related proteins whose exact form varies with the position in the cochlea.

mRNA messenger RNA

spliceosomes RNA-protein complexes that remove introns from RNA transcripts.

times as many as the much simpler fruit fly, *Drosophila melanogaster*. How can the much greater size and complexity of humans be encoded in only twice the number of genes required by a fly? The answer to this paradox is not fully understood, but it appears that humans and other mammals may be more adept than other organisms at encoding many different proteins from each gene. One way they do this is through alternative splicing, the processing of a single RNA transcript to generate more than one type of protein.

In most eukaryotic genes, the protein-coding sequences, termed exons, are interrupted by stretches of sequence, termed introns, that have no protein-coding information. After the gene is copied, or transcribed, to RNA, the introns are removed from this “pre-mRNA,” and the exons are spliced together to form a mature **mRNA**, consisting of one contiguous protein-coding sequence. In addition, the complete mRNA contains upstream and downstream sequences flanking the coding sequences. These sequences do not encode protein, but help to regulate translation of the mRNA into protein. Variations in the splice pattern lead to alternative transcripts and alternative proteins.

Splicing is accomplished in the cell’s nucleus by **spliceosomes**, which are molecular machines composed of proteins and small RNA molecules. The boundaries between exons and introns in a pre-mRNA are marked very subtly. Certain segments of the pre-mRNA, termed splice sites, direct the spliceosomes to the precise positions in the transcript where they can excise introns and splice together exons. Splice sites are short sequences, typically less than ten bases long. 5’ splice sites mark the 5’ end of introns; 3’ splice sites define the 3’ end of introns. (“Five prime” and “three prime” refer to the upstream and downstream ends of the RNA.)

Although splice sites often can be recognized as such by common patterns in their base sequence, there are many variations on the basic splice

site consensus sequence. These differences affect how readily a particular splice site is recognized and processed by the splicing machinery. Many other molecules within the cell, called splicing factors, also participate in the splicing reaction. The combination of all of these determines the pattern of splicing for a particular pre-mRNA molecule.

For many genes the pattern of splicing is always the same. These genes encode many copies of their corresponding pre-mRNA molecules. The introns are removed in a consistent pattern, producing mature mRNA molecules of identical sequence, all of which encode identical proteins.

For other genes the splice pattern varies depending on the tissue in which the gene is expressed, or the stage of development the organism is in. Because the choice of splice sites depends on so many different factors, the same pre-mRNAs from these genes may become spliced into several, or even many, different mature mRNA variants. 5' splice sites may be ignored, converting intron sequences into exons; 3' splice sequences can be ignored, converting exon sequences into introns; or different sequences, ordinarily not recognized as splice sites, can function as new splice sites. (To understand why ignoring a 5' splice site would convert an intron to an exon, recall that transcription of RNA proceeds from 5' to 3'.) The production of such mRNA variations through the use of different sets of splice sites is known as alternative splicing. It has been estimated that at least one-third of all human genes are alternatively spliced.

Alternative splicing can have profound effects on the structure and function of the protein encoded by a gene. Many proteins are comprised of several domains, or modules, that serve a particular function. For example, one domain may help the protein bind to another protein, while another domain gives the protein enzymatic activity. By alternative splicing, exons, and, therefore, protein domains, can be mixed and matched, altering the nature of the protein. By regulating which splice patterns occur in which tissue types, an organism can fine-tune the action of a single gene so it can perform many different roles.

The various forms of a protein are known as isoforms. Isoforms are often tissue-specific. The dystrophin gene, for example, has one form in muscle and another in brain tissue. Defects in alternative splicing are associated with several important human diseases, including amyotrophic lateral sclerosis, dementia, and certain cancers.

Alternative splicing can also act to turn genes off or on. In mRNA, codons, consisting of three adjacent **nucleotides**, either encode an amino acid or signal the ribosome to stop synthesizing a **polypeptide**. Normally, exon sequences must not encode stop codons (AUG, UAG, or UAA) until after the final amino-acid-coding codon. Alternative splicing can introduce a stop codon in the beginning or middle of a protein-coding sequence, resulting in an mRNA that encodes a prematurely truncated polypeptide.

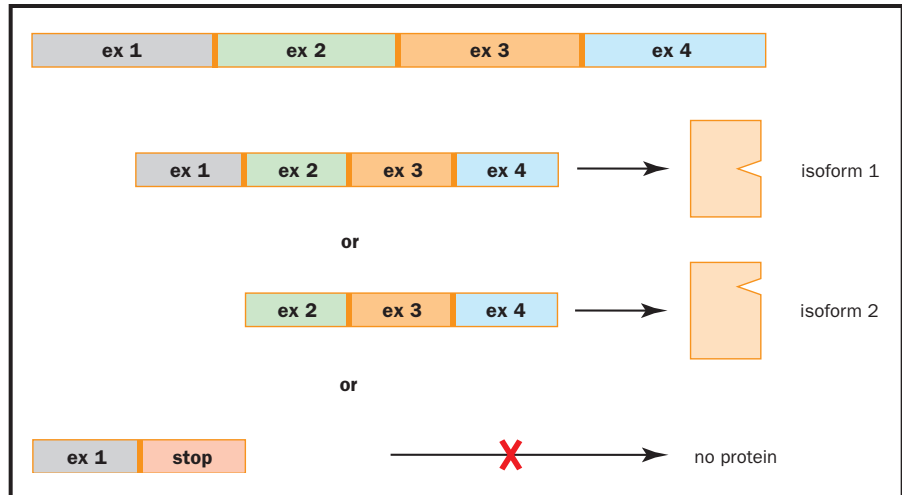
Human hearing offers a dramatic illustration of how important alternative splicing is in everyday life. Microscopic hair cells lining the inner ear vibrate when stimulated by sound. One of the proteins in the hair cells that plays a role in the hearing sensation is a calcium-activated potassium channel. The gene for this protein can generate more than five hundred different mRNA variants through alternative splicing. The resulting potassium-

The protozoan *Trypanosome brucei*, which causes African sleeping sickness, edits some of its messenger RNA molecules after they are transcribed. Uracil nucleotides are added in some locations in the mature RNA and deleted from others. Similar cases of RNA editing occur in other organisms, and even in humans. The human apolipoprotein B gene is edited in the intestine but not in the liver, leading to two distinct forms of the protein, serving different functions in the two organs.

nucleotides the building blocks of RNA or DNA

polypeptide chain of amino acids

Alternative splicing creates protein isoforms or may lead to no protein production.



channel proteins have slightly differing physiological properties. This is in part what tunes hair cells to different frequencies. SEE ALSO GENE; PROTEINS; RNA PROCESSING; TRANSCRIPTION.

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Alzheimer's Disease

Alzheimer's disease (AD) is a diagnosis applied to a group of degenerative brain disorders with similar clinical and pathological characteristics. It is the most common cause of dementia, with onset of symptoms after the age of fifty-five years. It is recognized as a major public health concern in societies with an aging population. AD affects four million people in the United States. At least 90 percent of those affected are over sixty-five years of age. In 1998 direct health care costs were estimated to be \$50 billion. Indirect costs, such as lost productivity and absences from work, were estimated to be \$33 billion.

First Description of AD

In 1907, Alois Alzheimer, a German physician from Bavaria, published the case of one of his patients. The patient, Mrs. Auguste D., at the age of fifty-one years developed an unfounded jealousy regarding her husband. This behavioral change was followed closely by a subtle and slow decline in other cognitive abilities, including memory, orientation to time and to physical location, language, and the ability to perform learned behaviors. All of her difficulties gradually progressed in severity. Within three years, the patient did not recognize her family or herself, could not maintain her self-care,



Eugenie Bonenfant, left, is a resident in Rhode Island's first assisted living community designed exclusively for people with Alzheimer's disease. The unit supervisor, Margaret Knight, visits, and she is surrounded by her own, familiar furniture.

and was institutionalized. She died a short four and a half years after her illness began. Her brain was removed at autopsy. Using a novel (at the time) silver stain to highlight changes in brain sections, Dr. Alzheimer viewed the tissue under his microscope. He described what are now the pathologic lesions of the disease that bears his name: loss of **neurons**, senile plaques found in the brain substance but outside of the neurons, and neurofibrillary tangles found inside neurons.

Dr. Alzheimer's patient had developed dementia. Dementia is an acquired and continuing loss of thinking abilities in three or more areas of cognition (which include memory, language, orientation, calculation, judgment, personality, and other functions) severe enough that the individual can no longer function independently at work or in society. There is no decrease in level of consciousness. Early in the illness, physical strength is maintained, though later the individual may "forget" how to perform certain physical functions, such as using tools or utensils, dressing, or performing personal hygiene activities. Onset of dementia may occur over days, months, or years. Its course may be static or progressive. Causes of dementia, other than AD, include other neurodegenerative disease, central nervous system infection, brain tumor, metabolic disease, vitamin deficiency, and **cerebrovascular disease**.

An Evolving Understanding of Dementia

Within three years of the publication of Dr. Alzheimer's first case, the term "Alzheimer's disease" was applied to patients who developed significant difficulty in memory and other areas of cognition at an age less than sixty-five years. Individuals who developed such symptoms later in life, generally after the age of sixty-five, were said to be suffering from senility, a process

neurons brain cells

cerebrovascular disease
stroke, aneurysm, or
other circulatory disorder
affecting the brain



considered a normal part of aging. The phrase “hardening of the arteries,” implying narrowing of arterial size with a reduction in blood flow to the brain, was used by physicians and by laypersons to designate the reason for senility. However, a causal relationship between arterial narrowing and senility had not been established scientifically.

Critical research reports were published in 1968 and 1970 providing evidence that senility and the disease Alzheimer described were similar both clinically and pathologically. Patients in each category developed similar and multiple cognitive deficits. Patients in each category developed plaques and tangles, and the majority of those diagnosed with senility did not have evidence of “hardening of the arteries.” Over the next decade senile dementia, Alzheimer's type, would replace senility as the accepted common cause of late-life dementia.

In 1984, consensus criteria for a clinical diagnosis of AD were established. Cardinal features include the insidious onset of decline in at least two areas of cognition, gradual progression of severity in these spheres resulting in dementia, onset of symptoms between the ages of forty and ninety years (most often after age sixty-five), and absence of another medical condition that by itself could cause dementia. **Pathological** study of tissue after death should reveal the characteristic findings of senile plaques in age-associated numbers (numbers larger than expected for the individual's age) and of neurofibrillary tangles. Using these criteria, both Alzheimer's disease as a presenile disorder and senile dementia, Alzheimer type, are subsumed into the broader diagnosis, Alzheimer's disease.

Genetics of Alzheimer's Disease

There are three areas of evidence that indicate a genetic basis for AD. First, it occurs as a Mendelian, autosomal dominant disease of early onset (occurring before the age of sixty) in multiple families. However, the number of such families with autosomal dominant inheritance is small. Second, it is generally the case that if an individual has a first-degree relative (parent or sibling) with AD, he or she has a greater risk of developing the disease than a person with no affected first-degree relative. Finally, AD is more likely to occur in each of a pair of **identical twins** than it is to occur in a pair of **fraternal twins**.

Recognizing these observations, in the mid-1980s researchers initiated scientific efforts to identify genes of importance in the disease, using the then-emerging recombinant DNA technology. By 1995, three causative genes and one susceptibility gene had been identified: *APP*, *PS1/2*, and *APOE*.

APP. In 1991, a British research group identified mutations in the APP gene that occurred only in patients with AD in very rare families. (Less than twenty such families have been reported in the medical literature.) The mutations were not found in family members who did not have AD. The APP gene codes for amyloid precursor protein, one of whose degradation products is a main constituent of the senile plaques of AD.

PS1 and PS2. In 1992, using **linkage analysis** of data from early-onset, autosomal-dominant families, researchers in Seattle, Washington; Jacksonville, Florida; and Antwerp, Belgium, almost simultaneously determined

pathological altered or changed by disease

identical twins monozygotic twins who share 100 percent of their genetic material

fraternal twins dizygotic twins who share 50 percent of their genetic material

linkage analysis examination of co-inheritance of disease and DNA markers, used to locate disease genes

GENES FOR ALZHEIMER'S DISEASE

Age at Onset	Inheritance	Chromosome	Gene	Protein	% AD
Early Onset	AD	14	<i>PS1</i>	presenilin 1	< 2
Early Onset	AD	21	<i>APP</i>	amyloid precursor protein	< 20 families*
Early Onset	AD	1	<i>PS2</i>	presenilin 2	3 families*
Early Onset	AD	?	?	?	?
Late Onset	Familial/ Sporadic	19	<i>APOE</i>	apolipoprotein E	~50
Late Onset	Familial	12p11-q13	?	?	?
Late Onset	Familial	9p22.1	?	?	?
Late Onset	Familial	10q24	?	?	?
Late Onset	?	?	?	?	?

Age of Onset: Early Onset: < 60 years, late onset: > 60 years; **Inheritance:** AD: autosomal dominant, familial: disease in at least one first-degree relative, sporadic: disease in no other family member; **Chromosome:** number, arm, and region; **Gene:** designation of identified gene; **Protein:** name of protein coded for by the gene; **% AD:** percent of AD caused by or * number of families identified with AD for each gene.

that a then-unknown gene for early-onset AD was located on chromosome 14. In 1995, a research scientist in Toronto, Canada, identified this gene as *PS1*, which codes for the protein called presenilin1. Individuals who have mutations in the gene consistently develop AD. Also in 1995, using comparative genomic techniques, the Seattle research group cited above identified the *PS2* gene, which codes for the protein termed presenilin 2. Using data from a few large, genetically isolated families with early- and late-onset disease, they determined that mutations in the gene consistently occur only in patients with AD.

APP, *PS1*, and *PS2* are causative genes: When mutated, each causes AD. If a person has a mutated gene, he or she will develop the disease at about the same age as others who have the same mutation. The risk of developing the disease approaches 100 percent.

APOE. In 1993 researchers in Durham, North Carolina, reported that one form (**allele**) of the *APOE* gene occurred more commonly in patients with late onset AD than was expected given its occurrence in the population as a whole. Numerous additional research groups corroborated the finding. The *APOE* gene occurs in three forms (alleles), determined by the DNA sequence. The three forms are termed *APOE*ε2, *APOE*ε3, and *APOE*ε4, and they code for apolipoprotein E molecules differing from one another by only one or two amino acids. APOE is a susceptibility gene; it imparts an increased risk of disease occurrence but by itself does not cause the disease. The presence of the ε4 form (*APOE*ε4) in either one or two copies in an individual increases the likelihood that the individual will develop AD. Occurrence may depend on other genetic factors or environmental factors or some combination from each category.

Additional families exist with early-onset, autosomal-dominant AD with no *APP*, *PS1*, or *PS2* mutations. Such families provide evidence that there may be additional causative genes. Whole-genome-scan analyses reported in the late 1990s provide evidence of additional susceptibility genes on chromosomes 9, 10, and 12. The genes located on these chromosomes have yet to be identified.

allele particular form of genes

Rationale for a Genetic Approach to Alzheimer's Disease

Alzheimer's disease, broadly defined, is a complex genetic disorder: Multiple causative and susceptibility genes acting singly or in concert produce similar symptoms and pathologic changes in patients. In each of its forms, it manifests age-dependent penetrance, meaning that the older an individual becomes, the more likely it is that he or she will develop the disease. Disease manifestations (such as age of onset or rate of progression) may be influenced by environmental exposures (alcohol use, head injury) or other health conditions (such as cerebrovascular disease). Identification of AD genes will lead to a better understanding of the cellular processes that cause dementia.

Currently, amyloid production from amyloid precursor protein is the focus of much research, although debate continues about its role. Amyloid production and deposition in the brain are affected by each of the four known AD genes. Decrease in amyloid production or increase in amyloid metabolism with a resulting decrease in deposition may result in delayed age of onset or slower progression of disease. Thus, alteration of amyloid processing of sufficient magnitude might result in disease prevention. Once process-altering treatments become available, knowing who is at risk for the disease will be important.

Genetic Testing and Alzheimer's Disease

DNA testing can be performed to determine whether an individual has a mutation in one of the causative genes and/or whether he or she carries one or two copies of the *APOEε4* susceptibility gene. Whether to test and which test to perform will depend on three conditions: family history of dementia, age of onset of disease, and clinical status of the individual. If a person has dementia, the test result could be useful in determining that the cause of the dementia is a form of AD. If a person has no symptoms of dementia, an estimate of the individual's risk could be developed, using the test. In the case of such estimates, both the actual accuracy of the test and the tested individual's understanding of its accuracy are of concern. While the consensus is that presymptomatic testing for causative mutations may be performed with appropriate counseling, debate over the safety and utility of *APOE* testing for individuals who do not show symptoms of Alzheimer's is ongoing.

In 2001, there was no treatment that prevented, much less cured, AD. Information regarding the risk of developing AD is useful only in life planning activities (such as purchasing or offering health insurance coverage or long-term care insurance coverage, or choosing retirement age) or in family planning. An individual's ability to cope with either an increased or a decreased risk may vary. Misuse of the information resulting in insurance or employment discrimination is possible. Absence of a causative gene mutation or of an *APOEε4* susceptibility gene in either symptomatic or presymptomatic disease does not preclude AD as the cause of dementia or mean that the individual has no risk of developing AD in later years. SEE ALSO COMPLEX TRAITS; DISEASE, GENETICS OF; GENE DISCOVERY; GENETIC TESTING; INHERITANCE PATTERNS; PSYCHIATRIC DISORDERS.

P. C. Gaskell Jr.

The first preimplantation testing for the APP mutation was announced in February 2002. Four gene-negative embryos from a gene-positive woman were selected and implanted, and she gave birth to one child who was free of the gene mutation, which causes early-onset Alzheimer's disease.

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Ames Test

The Ames test is a **protocol** for identifying mutagenic chemical and physical agents. Mutagens generate changes in DNA. Many mutagenic agents modify the chemical structure of adenine, thymine, guanine, and cytosine, the bases in DNA, changing their base-pairing properties and causing mutations to accumulate during DNA synthesis.

Ethyl methanesulfonate (EMS), for example, is a very potent mutagen. The ethyl group of EMS reacts with guanine in DNA, forming the abnormal base O⁶-ethylguanine. During DNA **replication**, DNA **polymerases** that **catalyze** the process frequently place thymine, instead of cytosine, opposite O⁶-ethylguanine. Following subsequent rounds of replication, the original G:C base pair can become an A:T pair. This changes the genetic information, is often harmful to cells, and can result in disease. Many mutagens cause a wide variety of cancers in humans.

During the 1960s the biologist Bruce Ames developed a test that still carries his name and that is still used as a relatively inexpensive way to assess the mutagenic potential of many chemical compounds. The procedure uses the bacteria *Salmonella typhimurium*. Wild-type *S. typhimurium* grows well on agar that contains only minimal nutrients. It can thrive on agar that contains only sugar, ammonium salts, phosphate, sulfate, and some trace metal ions. Amino acids are not needed because the bacteria have genes that encode enzymes that can make all twenty amino acids.

Ames developed strains of *S. typhimurium* that contain mutations in genes that the bacteria use to make the amino acid histidine. Such *his*[−] strains cannot survive unless histidine is added to their agar. Ames reasoned that mutagenic agents could cause changes in the aberrant gene that encodes the defective *his*[−] enzyme, causing it to revert back to the normal form, encoding the active protein. (The mutagen would likely also cause many other, undetected mutations.) A mutation that returns a function to a mutant is called a reverse mutation. The Ames test measures the ability of *his*[−] *S. typhimurium* to grow on agar that does not contain histidine. Growth indicates that a reverse mutation has reverted the *his*[−] gene back to an active form.

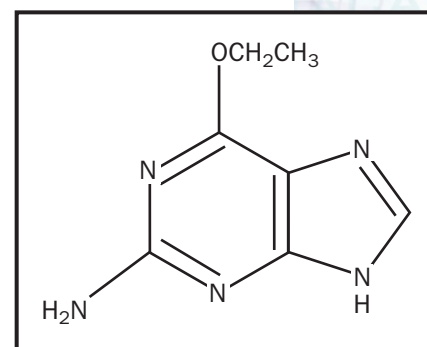
A typical Ames test involves exposing *his*[−] *S. typhimurium* to a test agent and then placing the exposed bacteria in petri dishes that contain agar with

protocol laboratory procedure

replication duplication of DNA

polymerases enzyme complexes that synthesize DNA or RNA from individual nucleotides

catalyze aid in the reaction of



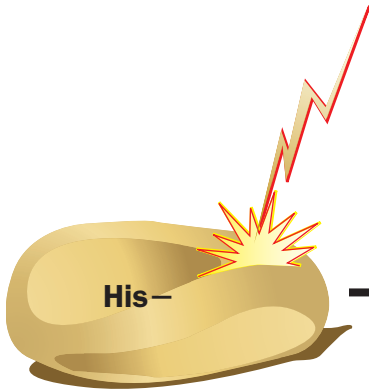
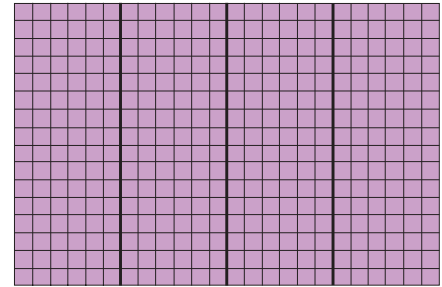
O⁶-ethylguanine.

AMES II ASSAY

Ames II *Salmonella*
unable to make histidine



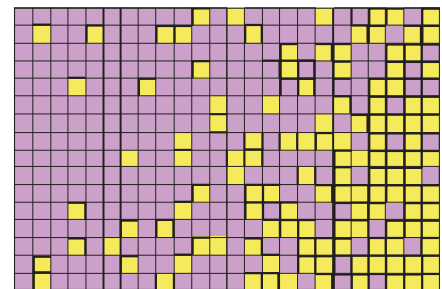
Medium without histidine



DNA damaging agent



Some *Salmonella* regain
ability to make histidine



Modern robotic technology is used to perform the Ames test today, but the fundamental procedure remains the same. Each square represents a single well in a plastic growth plate. Mutated *Salmonella* (yellow) are recorded by computer-driven detectors. Courtesy of Xenometrix, Inc., a subsidiary of Discovery Partners International, Inc.

incubating heating to optimal temperature for growth

no histidine. After **incubating** the dishes, the bacteria that have grown are counted. This number, which reflects the bacteria that undergo a reverse mutation from *his⁻* to *his⁺* *S. typhimurium*, is compared to the number of bacteria that undergo reverse mutations when they are not exposed to the agent. If the agent causes too many reverse mutations above those measured as spontaneous, it is considered to be mutagenic.

The Ames test can detect mutagens that work directly to alter DNA. In humans, however, many chemicals are *promutagens*, agents that must be activated to become true mutagens. Activation, involving a chemical modification, often occurs in the liver as a consequence of normal liver activity on unusual substances. Bacteria such as *S. typhimurium* do not produce the enzymes required to activate promutagens, so promutagens would not be detected by the Ames test unless they were first activated. An important part of the Ames test also involves mixing the test compound with enzymes from rodent liver that convert promutagens into active mutagens. These potentially activated promutagens are then used in the Ames test. If the liver enzymes convert the agent to a mutagen, the Ames test will detect it, and it will be labeled as a promutagenic agent.

The Ames test is widely used by the pharmaceutical industry to test drugs prior to using them in clinical trials. When a drug is mutagenic in the Ames test, it is usually rejected for further development and will probably not be tested in animals or used therapeutically in humans. The cosmetic industry also uses the Ames test to assess the mutagenic potential of makeup

and other hygienic products. The Food and Drug Administration requires companies to perform the Ames test before marketing most drugs or cosmetics. SEE ALSO CANCER; CARCINOGENS; MUTAGEN; MUTAGENESIS; MUTATION; NUCLEOTIDE.

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Androgen Insensitivity Syndrome

Androgen insensitivity syndrome (AIS) is a disorder caused by mutation of the gene for the **androgen** receptor. This protein binds testosterone and regulates the expression of other genes that stimulate male sexual development. Testosterone is the principal male androgen. AIS is an X-linked recessive disorder that completely or partially prevents development of male sexual characteristics despite the presence of the Y chromosome. Thus, the **phenotype** of a person with AIS, typified by female or ambiguous sexual characteristics, is at odds with the **genotype**, which includes the presence of both the X and Y, or male-determining, chromosomes.

The extent of the syndrome ranges from complete androgen insensitivity and development of normal external (but not internal) female sexual anatomy, to partial insensitivity, with altered or ambiguous male or female genitals, to mild insensitivity, with normal male genitals, enlarged breasts, and possibly impotence. Treatments depend on the extent of the syndrome, and may include hormone therapy, surgery, and psychological counseling. Gene testing and genetic counseling are available for families with affected members.

Sexual Development

AIS can best be understood against the background of normal human sexual development, which begins in the womb. The **gonads** arise from the same embryonic tissue, which is differentiated into one or the other by the actions of several genes not involved in AIS. In males, the most important gene is *SRY*, located on the Y chromosome. When present, this causes testis development. The genes responsible for ovary development are not as well characterized.

Once differentiated, the ovaries produce estrogen, and testes produce testosterone. These two hormones provide crucial signals for the differentiation of other sex-related characteristics, including an important set of primitive ducts.

The tubes and cavities that will house the adult's eggs or sperm after they leave the gonads develop from two different sets of ducts. Early in development, every fetus has both sets of ducts. One set, called the Wolffian ducts, has the capacity to develop into the male vas deferens and accessory

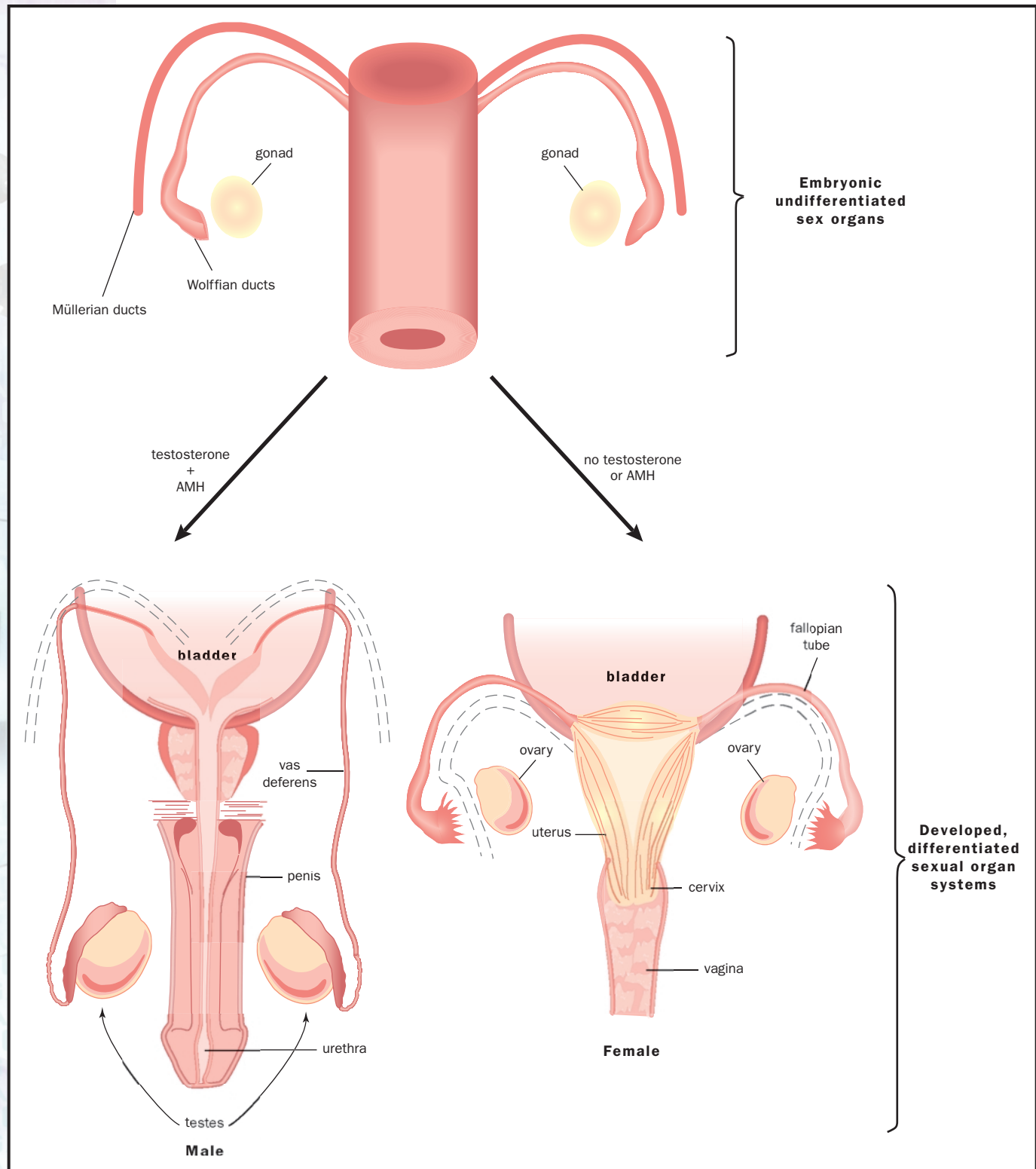
androgen testosterone or other masculinizing hormone

phenotype observable characteristics of an organism

genotype set of genes present

gonads testes or ovaries





Normal sexual development is governed by presence or absence of testosterone and AMH. In males, degeneration of the Mullerian ducts is triggered by AMH. Development of the Wolffian ducts is triggered by testosterone. In individuals with androgen insensitivity syndrome, AMH acts but testosterone does not.

structures, which store, nourish, and ejaculate sperm. The other set, called the Müllerian ducts, has the capacity to become the female fallopian tubes, uterus, cervix, and upper vagina.

In males, testosterone from the testes stimulates the development of the Wolffian ducts. Testosterone also stimulates nearby tissue to swell and form the penis and scrotum. A second hormone made by the testes, called anti-Müllerian hormone (AMH), induces the Müllerian ducts to undergo **apoptosis**, causing them to degenerate. During puberty, testosterone stimulates the development of other male secondary sex characteristics, including facial hair and a deepening of the voice.

In females, the absence of testosterone and AMH causes the Müllerian ducts to develop and the Wolffian ducts to degenerate. The same tissue that forms the penis and scrotum in males forms the clitoris, labia, and lower vagina in females. At puberty, estrogen stimulates development of female secondary sex characteristics, including enlargement of the breasts and onset of menstruation.

Testosterone and Its Receptor

Testosterone is a hormone, a molecule released in one set of cells that regulates the action of other cells. Testosterone exerts its action on these target cells by first binding with a receptor, called the androgen receptor (AR). The receptor is a protein that resides within the target cell. The testosterone-receptor complex moves to the nucleus of the target cell, where it acts as a transcription factor. **Transcription factors** bind to DNA to control the rate of gene expression. In the case of testosterone, the genes affected are those that “masculinize” the fetus, triggering the transformation of the Wolffian ducts into the mature male sexual anatomy and causing other, more subtle changes, including in the brain. Thus, the interaction of testosterone with its receptor is the principal means by which the male genotype (presence of a Y chromosome) leads to the development of the male phenotype (presence of a vas deferens, penis, and accessory structures).

The Consequences of Androgen Insensitivity

With an understanding of normal sexual development, consider the consequences of complete androgen insensitivity on the events of development in a person with the XY genotype. Since the Y chromosome is present, there will be testes. The testes will produce testosterone and AMH. AMH will cause degeneration of the Müllerian ducts, and so there will be no fallopian tubes, uterus, cervix, or upper vagina. But the defective receptor means that testosterone cannot exert its effects, so the Wolffian ducts also degenerate, and there will be no vas deferens. There will also be no penis or scrotum. Instead, the testes remain in the abdomen (where they originate), and the exterior tissue develops a short vagina that ends in a blind pocket. In milder forms of the syndrome, with only partial insensitivity to androgens, the genital structures may be ambiguous, with varying degrees of male versus female predominance.

The Androgen Receptor Gene and Protein

The AR gene is located on the long arm of the X chromosome, at a location that is designated as Xq11-q12. The gene is about 90,000 nucleotides

apoptosis programmed cell death

transcription factors proteins that increase the rate of gene transcription

